

Design and implementation of data acquisition system based on LabVIEW

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ABSTRACT

With the development of electronic information technology in the 21st century, various testing and acquisition techniques have made breakthrough progress. However, with the rapid development of commercial society and economy, traditional acquisition methods no longer meet current requirements, especially in the measurement industry. LabVIEW, with its intuitive visual programming interface, rich functional modules, and compatibility with other software, has become one of the preferred tools for engineers to perform data acquisition, analysis, and control. Users can build a data acquisition system that suits their practical applications using LabVIEW, and can monitor and record data from various sensors and instruments in real-time. Furthermore, the collected data can be analyzed and processed. In this paper, a weak current acquisition device is designed using the USB6001 data acquisition card as an example.

Keywords: LabVIEW; virtual instrument; data acquisition system.

1. INTRODUCTION

LabVIEW has attracted more and more attention and has been extensively applied in day-to-day work and scientific research. Unlike traditional coding, it presents programs in VI icons and wiring diagrams, enabling designers and learners to quickly understand the logic and observe data transmission processes. As one of the most widely used software globally, LabVIEW is a powerful tool that can perform various tasks, including numerical computation, signal processing, data acquisition, image acquisition, and transmission. Additionally, Matlab algorithms can be invoked on LabVIEW, making it a comprehensive solution. This paper proposes a voltage data acquisition system based on LabVIEW, which can display collected waveforms and save data^[1]. LabVIEW is widely used in China for automation control, instrument measurement, and data acquisition and processing. Liu et al. (2021) utilized LabVIEW as the foundation for designing a remote monitoring system for greenhouse cultivation. Zhang et al. (2022) developed a data acquisition and processing system for monitoring transformer vibration signals using a client-server architecture based on LabVIEW. Internationally, LabVIEW finds even broader applications. Zhang et al. (2020) designed and constructed a remote indoor environmental monitoring system with ZigBee technology, incorporating a user monitoring platform developed in LabVIEW for real-time display, data storage, and threshold alerts. Bai et al. (2017) employed LabVIEW to monitor the environmental comfort of laying hens in real-time, utilizing fuzzy comprehensive evaluation theory for data fusion of environmental factors and applying the Analytic Hierarchy Process (AHP) to determine the weights of environmental factors during winter and summer seasons. A real-time data collection and analysis system for housing environment was built on the LabVIEW platform, ultimately developing a novel method called the Comfort Time Ratio for statistical data collection on environmental comfort during the feeding period of laying hens. In summary, LabVIEW is extensively used in China for automation control, instrument measurement, and data acquisition and processing. It has been applied in various fields such as greenhouse cultivation, transformer monitoring, and environmental monitoring systems. Internationally, LabVIEW's applications include remote monitoring, environmental assessment, and real-time data collection and analysis in different domains.

2. SYSTEM COMPOSITION AND ARCHITECTURE DESIGN

The data acquisition system, designed using virtual instrument technology, consists of two main components: hardware circuitry and acquisition programs. The acquisition programs are primarily developed by engineers and designers using LabVIEW's visual programming language^[2]. This allows for the creation of a graphical user interface (GUI) for operation and data processing. The hardware component includes sensors, excitation circuits, amplification circuits, etc. The data acquisition card, such as NI's USB6001, is compatible with LabVIEW. Among the sensors, the magnetic flux gate sensor

is preferred for its strong anti-interference capability, high accuracy, and cost-effectiveness. It only requires a core and a coil, making it the primary choice for weak current measurements [3]. This software can run waveform output, data acquisition, and data processing simultaneously, such as using Fast Fourier Transform (FFT) or interpolation. It adopts a large-scale multitasking structure and fully utilizes the multi-core and multi-thread performance of the computer CPU to meet experimental requirements. Due to complex data analysis requiring integration with professional analysis platforms, each data acquisition and analysis requires programming from scratch. Therefore, this paper focuses on designing a universal software platform, emphasizing the system's hardware and software versatility and expandability, to reduce repetitive work. The system composition is illustrated in Figure 1.

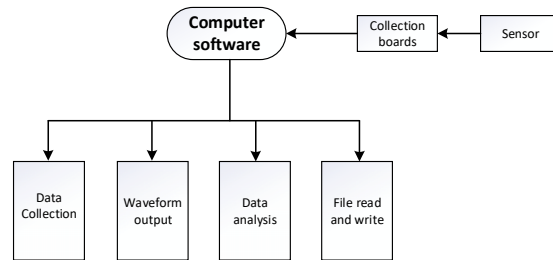


Figure 1. System composition and architecture.

3. SYSTEM DESIGN

The system design should be user-oriented, considering the needs of users and keeping up with the pace of the times. Therefore, we have designed a system that provides a friendly visual interface. To ensure ease of use and a good user experience, the visual interface is divided into several functional modules: the user login interface for identity authentication and access control, the user registration interface for new user registration and account management, the user management interface for managing the permissions and information of registered users, the real-time data acquisition interface for displaying the collected data in real-time, and the data analysis and processing interface for further analyzing and processing the collected data^[4]. These interfaces are designed to provide a clear and concise operating environment, allowing users to easily use the system and obtain the necessary data at any time.

3.1 Database selection and utilization

In most database systems, data is organized through tables. Therefore, it is necessary to first define the structure of the table, including the names of each field, the data types used, precision, and other information. This process can be achieved in LabVIEW by using relevant functions, which are included in the LabVIEW Database Connectivity Toolkit. By using these functions, users can create data tables by writing code, which allows them to flexibly customize the database structure according to their needs. During the creation of data tables, users need to specify the name and data type of each field, as well as corresponding precision and other attributes^[5]. These detailed information will determine the data type and format of each column in the table.

After successfully creating the data table, the system can perform operations such as adding, deleting, querying, and updating data records through the established database connection. In addition to adding data records, other related DB Tools VIs can be used to perform delete, query, and update operations to meet different user needs. In this login system, the DB Tools Insert Data VI is used to update user login information^[6]. This design allows users to conveniently query and retrieve data from the database and easily display the results^[7].

3.2 Designing the login interface

Different levels of permissions allow individuals to perform different operations after logging in. Users with user-level permissions can modify their own information, select and read parameters, and export reports after completing experiments. Users with administrator-level permissions can add, delete, and modify users. The highest level, Debug, not only allows modification of user information but also modification of administrators.

When users first use the data collection system, they need to register their information. By filling out the registration form, users can create their own account and password, and their information will be saved in the database. When users click on the "Register" button in the login panel, the registration front panel will automatically open and the login system will be closed^[8]. If the entered password matches the confirmed password and the username is not empty, the system will check

if the duplicate username exists in the User Table. If it exists, the user will be prompted that the username is already registered. If it doesn't exist, a successful registration message will pop up, and the system will write the username and password into the User Table. If the entered password does not match the confirmed password or the username is empty, the system will prompt that the registration has failed. The user login interface is one of the key user interfaces in the data collection system. It provides users with secure, independent, and reliable identity verification and access control, ensuring the integrity and confidentiality of user information and data. Additionally, the login interface can provide other security measures such as captcha and security questions to enhance the security and reliability of the login process. These measures effectively prevent security threats such as malicious attacks and password guessing, thereby improving the security and stability of the system. When users enter login information, the system will search the User Table to check if the information exists. If it does, the system will prompt a successful login and open the data collection interface. If the information does not exist, the system will prompt a login failure. The program flowchart is shown in Figure 2.

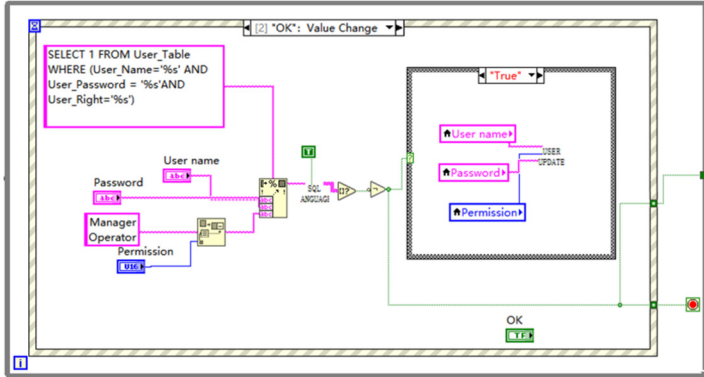


Figure 2. Flowchart of user login program.

The user management interface has functions for adding, deleting, and modifying users. It is aimed at administrators and Debug. When a user no longer needs to use the data collection system, the administrator can eliminate the user's account and password, and the system database will delete all data associated with that account. Additionally, if a user forgets their password, they can inform the administrator who can then modify or reset the password. The add function is mainly for Debug, which can modify user permissions [9]. The system performs a check to determine if the user has Debug permissions. If false, permission modification is disabled. If true, it is enabled. Only when the entered password is correct and the username is not empty, the system will find the corresponding username in the table and update the permission information in the table.

3.3 Data Collection and Storage

Data acquisition is achieved through a combination of the USB6001 data acquisition card and program, which measures weak currents using magnetic flux gate sensors and collects waveforms as shown in Figure 3.

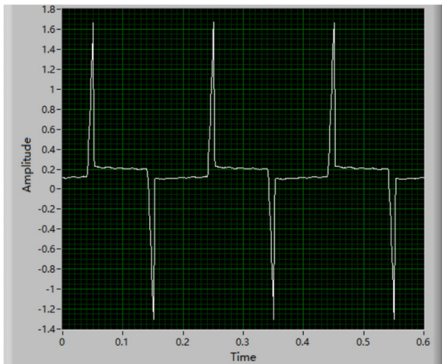


Figure 3. Magnetic core saturation current.

By using interpolation as a data processing method, the accuracy of the collected signals can be improved. During the processing, the collected signals are converted into digital format and saved in a table. Then, interpolation is applied to the

digital signals to enhance their accuracy. As clearly shown in Figure 4, the time precision is improved by a factor of 10 after the interpolation process. Subsequently, more accurate results can be obtained through corresponding formulas, thus demonstrating the integrity of the system.

12/5/2023	20:49:34.138739	-0.01786
12/5/2023	20:49:34.139739	-0.019149
12/5/2023	20:49:34.140739	-0.019149
12/5/2023	20:49:34.141739	-0.020438
12/5/2023	20:49:34.142739	-0.021727
12/5/2023	20:49:34.143739	-0.021727
12/5/2023	20:49:34.144739	-0.020438
12/5/2023	20:49:34.145739	-0.020438
12/5/2023	20:49:34.146739	-0.021727
12/5/2023	20:49:34.147739	-0.019149
12/5/2023	20:49:34.148739	-0.022942
12/5/2023	20:49:34.149739	-0.022942
12/5/2023	20:49:34.150739	-0.023282
12/5/2023	20:49:34.151739	-0.022993
12/5/2023	20:49:34.152739	-0.023704
12/5/2023	20:49:34.153739	-0.0247548
12/5/2023	20:49:34.154739	-0.0247548
12/5/2023	20:49:34.155739	-0.0258837
12/5/2023	20:49:34.156739	-0.0258837
12/5/2023	20:49:34.157739	-0.026487
12/6/2023	18:23:46.239953	-0.019738
12/6/2023	18:23:46.240053	-0.021542
12/6/2023	18:23:46.240153	-0.021834
12/6/2023	18:23:46.240253	-0.022854
12/6/2023	18:23:46.240353	-0.023273
12/6/2023	18:23:46.240453	-0.025023
12/6/2023	18:23:46.240553	-0.025935
12/6/2023	18:23:46.240653	-0.026263
12/6/2023	18:23:46.240753	-0.027548
12/6/2023	18:23:46.240853	-0.028351
12/6/2023	18:23:46.240953	-0.029732
12/6/2023	18:23:46.241053	-0.031143
12/6/2023	18:23:46.241153	-0.032601
12/6/2023	18:23:46.241253	-0.034108
12/6/2023	18:23:46.241353	-0.035529
12/6/2023	18:23:46.241453	-0.036907
12/6/2023	18:23:46.241453	-0.038529

Figure 4. Data processing result.

3.4 Data storage and representation

After importing waveforms into tables, we can visually observe the data, facilitating subsequent data processing. By presenting waveform data in tabular form, we can clearly see the corresponding values at each time point, enabling better understanding of the data's trends and characteristics. Furthermore, the structured representation of tables makes data manipulation more convenient. We can easily perform operations such as filtering, sorting, and aggregating to extract the desired information and conduct further analysis. Additionally, tables support various common mathematical and statistical functions, allowing for more precise data calculations and analysis. Therefore, importing waveforms into tables not only enhances data visualization but also provides a flexible and efficient means of data processing. This convenient data processing method helps us quickly obtain valuable information and supports deeper data analysis and decision-making [10].

4. CONCLUSION

This article introduces a highly scalable data acquisition system that combines software and hardware components. By configuring software functionality and determining software operational logic, efficient and stable data collection is achieved. The system focuses on measuring core saturation current and includes user-friendly registration and login interfaces. These interfaces facilitate convenient system usage and management, while the system itself possesses excellent expandability. Users can add collection programs based on their specific requirements to adapt to diverse data collection needs in various domains. Moreover, users can also add data analysis programs according to their specific requirements, enabling them to obtain accurate information and make informed decisions. The program enables signal triggering through real-time monitoring of collected signals. It can be applied to various types of acquisition cards and also provides expandability. Users can add new programs to the main program interface according to their needs. This system has been proven to be effective through experiments.

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